

Learning with less: label-efficient land cover classification at very high spatial resolution using semantic stratification and self-supervised deep learning

Author order?

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Abstract

Deep learning methods for land cover classification have been shown to be highly performant, particularly at high spatial resolutions where traditional pixel-wise or object-based approaches struggle. However, the wide parameter space of deep neural networks typically requires several thousand densely annotated imagery tiles to achieve reliable performance. This is problematic, as data annotation is costly and time-consuming, creating a strategic barrier to widespread adoption of such models for land cover classification at high resolutions. We address this challenge through two complementary approaches. First, we demonstrate that utilizing a stratification approach that groups candidate imagery tiles into semantic clusters using existing moderate resolution land cover products improves the quality of a training dataset compared to simple random sampling. Furthermore, applying Bootstrap Your Own Latent (BYOL) self-supervised learning to an ImageNet pre-trained ResNet-101 encoder using unannotated aerial imagery enhances the performance of a semantic segmentation network on a downstream land cover classification task, particularly when few samples are supplied during training. When applied to a 1 m resolution land cover task using linear probing, our BYOL model demonstrates average increases of 13.04% and 14.87% in mIoU and overall accuracy over a similar model pre-trained on ImageNet alone, illustrating the extent of the domain gap between general-purpose imagery often used for pre-training and high resolution remote sensing data. Using only 500 labeled samples during the training phase, our approach achieves 87.42% overall accuracy and 63.00% mIoU for comprehensive 8-class land cover mapping across Mississippi when the model is fine-tuned in a U-Net framework—improvements of 2.06% and 2.77% over ImageNet pre-training. These results demonstrate that creative approaches to sample stratification and selection along with advanced pre-training can substantially reduce the need for hand-annotated data, breaking down hurdles for large-scale high spatial resolution land cover mapping.

1 Introduction

Land cover data provides valuable information about the characteristics of the Earth’s surface for a wide variety of applications across many domains, including agriculture, urban planning, and environmental monitoring Ban et al. (2015); Loveland et al. (2000); Scarpace and Quirk (1980); Townshend (1992); Wulder et al. (2018). Research into automated methods for land cover classification from remotely sensed data has been ongoing for several decades, with a majority of methods relying on supervised learning techniques to classify land cover using multispectral aerial or satellite imagery Maxwell et al. (2018); Phiri and Morgenroth (2017). Historically, pixel-wise classification methods have been the predominant approach for land cover classification, where each pixel in the image is classified based on the pixel’s raw spectral values, hand-crafted features, and ancillary data from other geospatial datasets Khatami et al. (2016); McIver and Friedl (2001). This pixel-wise paradigm is effective for classifying coarse and moderate spatial resolution imagery (e.g., MODIS, Landsat, Sentinel-2), but is less effective for classifying high spatial resolution imagery due to decreased inter-class spectral separability and increased intra-class spectral variability Grekousis et al. (2015); Hansen and Loveland (2012); Lu and Weng (2007); Markham and Townshend (1981); Strahler et al. (1986); Woodcock and Strahler (1987). This is problematic, as ideally, high spatial resolution (< 10 m) land cover data offers more detailed information about the Earth’s surface with greater spatial accuracy, which is valuable for applications that require fine-grained information about land cover types and their spatial distribution Fisher et al. (2018); Li et al. (2022a). Object-based Image Analysis (OBIA) techniques have been proposed as a potential workflow to address these issues, but their performance is highly dependent on the quality of objects or superpixels extracted from the imagery during the segmentation process, which must be carefully tuned to the specific characteristics of the imagery and the land cover classes of interest Blaschke et al. (2014); Johnson and Xie (2011); Ma et al. (2017). Still, a key idea behind OBIA is that the spatial information contained in the objects can be leveraged to provide more information to the classifier, which addresses the inter-class spectral separability issues that hamper pixel-wise classification methods Blaschke et al. (2000); Myint et al. (2011); Tehrany et al. (2014). The application of deep computer vision models to land cover classification has been of significant interest in recent years, as these models excel at extracting spatial and spectral features from imagery in a data-driven manner, making them well-suited for high spatial resolution land cover classification tasks Ma et al. (2019); Vali et al. (2020); Zhu et al. (2017).

Convolutional neural networks (CNNs) are a class of deep learning models that have been widely used for land cover classification tasks due to their ability to automatically extract hierarchical spatial structures from the input imagery that are useful for classifying land cover type Kussul et al. (2017); Li et al. (2022b); Pan et al. (2020); Song et al. (2019). CNN-based semantic segmentation

models learn a mapping from the input image to a pixel-wise classification probability map, which can be used to generate a land cover map by assigning each pixel to the class with the highest probability Long et al. (2015). Semantic segmentation models have been widely used for land cover classification tasks, as they can facilitate the generation of high spatial resolution land cover maps that leverage both spectral and spatial information in the input imagery under a unified framework Yuan et al. (2021). However, these models require large amounts of labeled training data to learn the complex spatial and spectral patterns present in the imagery, which can be difficult to obtain for high spatial resolution imagery due to the high cost of collecting and labeling the data Hu et al. (2015).

This need for large amounts of ground truth data for accurate training of deep learning models has led to the extensive research into methods for reducing the amount of data necessary to train these models. Most of these methods focus on leveraging transfer learning, where the parameters of a model pre-trained using a large dataset are transferred to a model that is subsequently fine-tuned on a smaller dataset of interest. Transfer learning is effective for deep computer vision models, as the features learned by a model on a large dataset are often generalizable to other datasets thanks to the hierarchical nature of the features learned by the model Hu et al. (2015); Kolesnikov et al. (2020). However, the effectiveness of transfer learning is highly dependent on the similarity between the source and target datasets, as well as the amount of labeled data available for the target dataset. For example, a common approach to transfer learning for land cover classification tasks is to pre-train a model on a large general-purpose image classification dataset (e.g., ImageNet Russakovsky et al. (2015)) and then fine-tune the model on a smaller dataset of interest Pires de Lima and Marfurt (2020). This is generally effective, but the domain shift between the source and target datasets can hinder the performance of the model, and the extent of this domain shift between general purpose image classification datasets and high spatial resolution remote sensing datasets is not well understood Ma et al. (2024). Self-supervised pre-training methods have been proposed as an alternative to traditional fully supervised pre-training methods, as they can leverage the spatial and spectral information present in the input imagery to learn useful features through the use of pretext tasks that do not require labeled data Chen et al. (2020); Tian et al. (2020); Wu et al. (2018). In remote sensing, these methods are becoming more widely explored as a means to pre-train large foundation models that can serve as strong feature extraction backbones for downstream tasks **CITE!**. While effective, foundation models are geared for performance in multi-sensor, multi-modal, or cross-task scenarios, and not strictly label efficiency **CITE!**. As such, to the authors' knowledge, little work has been done on how these self-supervised pre-training approaches can impact operational land cover mapping under extreme low-data scenarios.

Beyond the choice of pre-training routine, another poorly understood component of developing semantic segmentation models for land cover classification is the choice of sampling routine used

to select training data. Ideally, a simple random sample of sufficient size from the region of interest should produce a quality training dataset. However, the sample size must be large enough to satisfy the conditions that the sampled data is representative of the variation in land cover across the region while also having enough samples in the minority classes such that a classifier can capture said classes accurately. Simple random sampling becomes problematic for large areas with diverse land cover types or strongly imbalanced land cover distributions. Much research on stratification strategies for building pixel-based datasets for land cover classification has been conducted to address the issues associated with simple random sampling. These stratification procedures involve splitting the population into several groups based on some intrinsic property (e.g., geospatial location) or information from an external data source (e.g., prior land cover information) and sampling from these groups proportionally. However, semantic segmentation models require small patches consisting of dense labels, that is, every pixel in the patch is assigned to a class, rather than individual pixels. While this does not negate the implications of stratification, it does add extra considerations to the process, as one patch can have multiple classes. For example, when stratifying training data by land cover class according to some prior knowledge of the land cover distribution, it can be very difficult to select patches in such a way that fit the desired land cover distribution. Further, the composition of classes in sampled patches should also be representative of spatial relationships between land cover types in the region of interest. **SEE** <https://chatgpt.com/c/6894f587-a918-832a-8112-73ab5926bb09>, <https://www.mdpi.com/2072-4292/15/6/1607#:~:text=Figure%205C%2C%20we%20show%20that,the%20reported%20confusion%20matrix%20accuracies> This issue is exacerbated under low-data scenarios, where it can be difficult to capture the full diversity of both land cover types and their spatial relationships with few samples.

In this work, we demonstrate that employing a novel stratification strategy that groups candidate patches according to their semantic content (i.e., land cover composition) using a coarse-resolution land cover dataset to guide the stratification selection such to sample patches that are representative of the diversity of land cover compositions found in the region of interest. Additionally, we employ Bootstrap Your Own Latent (BYOL) as a self-supervised pre-text task during pre-training, enabling a ResNet-101 convolutional encoder network to learn to extract deep features from aerial imagery without the need for full supervision using labelled data. This approach is used to tackle land cover classification at 1 m resolution over the state of Mississippi using only 1,000 patches selected using the semantic stratification procedure. We demonstrate that these approaches can be used to train deep semantic segmentation models that are accurate, generalizable, and capable of yielding acceptable performance even when supplied with only 250 patches of imagery-ground truth pairs during the supervised fine-tuning stage.

2 Related work

2.1 Approaches to very high spatial resolution land cover classification

maybe define VHSR in the intro???

VHSR land cover classification has historically been tricky due to blah blah blah Prior to deep learning, OBIA was the predominant paradigm - mention a few models Performance of OBIA at VHSR attributable to integration of spatial information into imagery classification Drawbacks, scale parameters, generalization, human intervention

Semantic segmentation introduced in computer vision community Why it worked well Prominence in VHSR classification today List a couple of notable works on convolution

2.2 Training data sampling strategies

Mention stratification, classical pixel-wise Struggles when applying to patch wise Mention a few

2.3 Remote sensing pre-training and self-supervised learning

Bring up pre-training, why its important, how transfer learning works Difficulty in remote sensing pre-training, unsure of domain shift between standard ImageNet pre-training and remote sensing imagery Emergence of self-supervised learning as a new paradigm for pre-training

How this has translated to remote sensing in the form of foundation models What these foundation models do, why they are built Problems with foundation models: ViT lacks inductive bias toward imagery Transformers still not the norm in remote sensing analysis, lack of inductive bias makes very low data fine-tuning difficult Still, can apply SSL to CNNs using contrastive/self-distillation approaches

3 Study area

Mississippi, USA The overarching goal of this proposed work is to develop a 1 m resolution land cover map for the state of Mississippi to facilitate high-fidelity land analysis. Mississippi is located in the southeastern United States, and is home to a diverse range of land cover types. The region's humid subtropical climate and fertile soils in the Mississippi River Delta make these areas home to a variety of agricultural activities Mississippi State University Extension, while the remainder of the state is characterized by forests and woodlands which comprise the majority of land cover in the state United States Geological Survey. The coastal region along the Gulf of Mexico is home to a sprawling urban network intertwined with herbaceous wetlands and barren beaches English

(2011). The state's diverse land cover types, geographical position along the Mississippi River and Gulf of Mexico, and economic dependence on agriculture and natural resources make the need for accurate land cover information at high spatial resolution a priority in order to support natural resource management, agricultural planning, and environmental monitoring efforts.

Virginia, USA In order to validate our proposed semantic stratification approach, we also investigate the performance of the proposed method using an existing land cover dataset. The Chesapeake Bay Program (CBP) land cover dataset is used to provide ground truth data for this phase of the study. The CBP land use and land cover products are among the largest meter-scale land cover datasets in the United States, and are produced by the Chesapeake Conservancy as part of a cooperative agreement between the US Geological Survey and the US Environmental Protection Agency Claggett et al. (2025). The 2024 edition provides a 56-class land use product at 1 m resolution over the Chesapeake Bay watershed, which includes portions of six states (Delaware, Maryland, New York, Pennsylvania, Virginia, and West Virginia) and the District of Columbia for the years 2013/2014, 2017/2018, and 2021/2022.

National Agriculture Imagery Program (NAIP) For this study, USDA National Agriculture Imagery Program (NAIP) imagery is used as the primary data source from which the models are trained and the land cover map is generated. NAIP imagery is high-resolution (0.6 meter) multispectral aerial imagery collected during the summer months over the contiguous United States. Such imagery is collected on a 2–3 year cycle (depending on the state) and is predominantly used for agricultural purposes. While the temporal availability of NAIP imagery is limited, the high spatial resolution of the imagery and public availability of the data make it an attractive data source for remote sensing applications that require high spatial resolution data over large areas in the contiguous United States Earth Resources Observation and Science (EROS) Center (2017). NAIP imagery collected in 2021 and 2023 for the states of Virginia and Mississippi, respectively, is used in this study. The imagery is provided in 4 spectral bands (Red, Green, Blue, and Near-Infrared) at a spatial resolution of 0.6 meters per pixel. The imagery is resampled to a spatial resolution of 1 m to reduce the computational cost of processing and storing the data, as well as to align with the desired spatial resolution of the land cover product we aim to generate. Mississippi imagery is reprojected to the Mississippi Transverse Mercator (MSTM) coordinate system (EPSG: 3813) in accordance with recommendations from the Mississippi Automated Resource Information System (MARIS) to ensure that all data is in the same coordinate reference system Mississippi Automated Resource Information System. Likewise, data for the Chesapeake Bay program is reprojected to CONUS Figure 1 shows the extent of the study area and the NAIP imagery used in this study in the MSTM coordinate system.

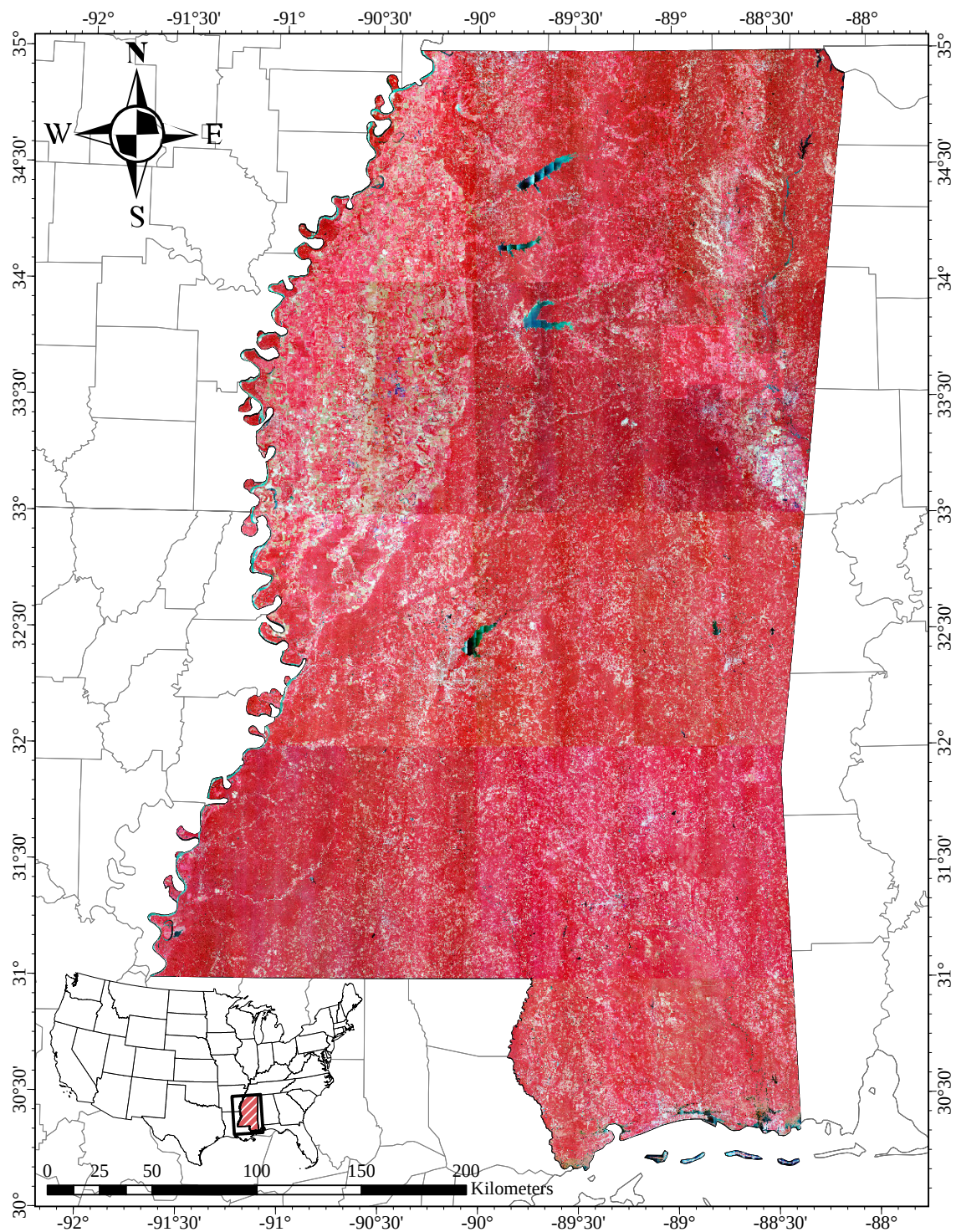


Figure 1: 2023 NAIP color-infrared imagery over Mississippi, USA.

4 Methodology

Our proposed methodology consists of a multi-stage approach to land cover classification that aims to produce a land cover product at high accuracy under data-scarce conditions using a machine-learning driven stratification protocol that aims to extract informative samples for training and the use of self-supervised learning to pre-train a large vision model for feature extraction from color-infrared aerial imagery in the absence of ground truth data. First, we apply the semantic stratification approach to train and evaluate land cover classification model using a subset of the CPB land use/land cover dataset in Virginia in order to validate the approach in the context of other stratification approaches. We then use this approach to select 1000 sample tiles over the state of Mississippi for manual visual annotation to build a training dataset. A separate, larger dataset of unannotated sample tiles from the state of Mississippi will be used to pre-train a ResNet-101 convolutional encoder using

5 Preliminary Results and Discussion

To collect preliminary results, 250 tiles from the full Mississippi land cover dataset were used for fine-tuning training, validation, and testing in the last stage of the multi-stage pre-training approach. In total, 119 tiles were used for training, 61 tiles were used for validation, and 70 tiles were used for testing. These early results give insight into the efficacy of this approach: Table 1 shows that introducing a denoising autoencoder pre-training stage prior to supervised pre-training has a substantial impact on a model’s performance. This is evidence that the core idea of the proposed approach has merit, in that the denoising autoencoder pre-training stage enables the model to learn spatial features that are useful for land cover classification, even when the model undergoes subsequent supervised pre-training on a different region. Additionally, the choice of which layers to freeze during the fully supervised pre-training stage has a substantial impact on the model’s performance. As shown in Table 2, the most effective strategy for freezing parameters is to train the full network during the fully-supervised pre-training stage, but freeze the encoder network during the final fine-tuning stage. Examining the results of the Chesapeake Bay Land Cover classification task (i.e., the second stage of the multi-stage pre-training approach) in Table 3 shows more meager results compared to the Mississippi Land Cover classification task, with the denoising autoencoder pre-training stage providing only a marginal improvement over ImageNet pre-training. This suggests that the denoising autoencoder pre-training stage is more effective when the model undergoes further pre-training in a supervised manner. Further, this opens the door to future work on using a stronger pre-training strategy for the first stage of the multi-stage pre-training approach, such as increasing the noise level in the denoising autoencoder or using a more complex

pre-text task to further improve the model’s performance.

Table 1: Performance comparison of land cover classification models on Mississippi test data showing how different pre-training strategies (random initialization, ImageNet weights, and ImageNet weights followed by denoising autoencoder pre-training) affect accuracy metrics when all models undergo the same two-stage fine-tuning process: full encoder training on Chesapeake Bay data followed by decoder-only fine-tuning on Mississippi data with frozen encoders.

Stage 1 Pre-Training Strategy	Overall Accuracy (%)	Overall F1 Score (%)	Macro F1 Score (%)	Loss
Random Initialization	28.96	30.47	10.34	36044
ImageNet	77.54	77.58	50.25	19638
ImageNet + Denoising Autoencoder	82.11	82.62	70.45	11364

Table 2: Effects of training on different portions of the network during the fully supervised training stages.

Weights Trained					Loss
Fully Supervised Pre-training Encoder	Decoder	Encoder	Decoder	Classifier Head	
	✓			✓	38126
	✓		✓	✓	19725
	✓	✓	✓	✓	14511
✓	✓			✓	38949
✓	✓		✓	✓	11364
✓	✓	✓	✓	✓	11725

Table 3: Performance metrics for models trained on the Chesapeake Bay Land Cover dataset with different pre-training strategies.

Pre-training strategy	Overall Accuracy (%)	Macro F1 Score (%)	Loss
None	77.51	52.59	11056
ImageNet	82.50	67.16	7393
ImageNet + Denoising Autoencoder	82.63	67.43	7439

The best model from the multi-stage pre-training approach was used to classify land cover in the state of Mississippi at 1 m resolution in order to understand the strengths and limitations of the proposed approach. A visual assessment of the land cover classification results for the Mississippi dataset is shown in Figure 2. The confusion matrix for the land cover classification results is shown in Table 5, and the classification report is shown in Table 5. These results show that, overall, the

model is fairly accurate, even with the limited labeled data available for training. However, more data is needed to improve the model's performance on certain classes, such as cultivated crops, barren land, and open water. This suggests that, despite the promising results of the multi-stage pre-training approach, more labeled data is needed to fully leverage the feature learning capabilities of the model and improve its performance on classes that are underrepresented in the training data.

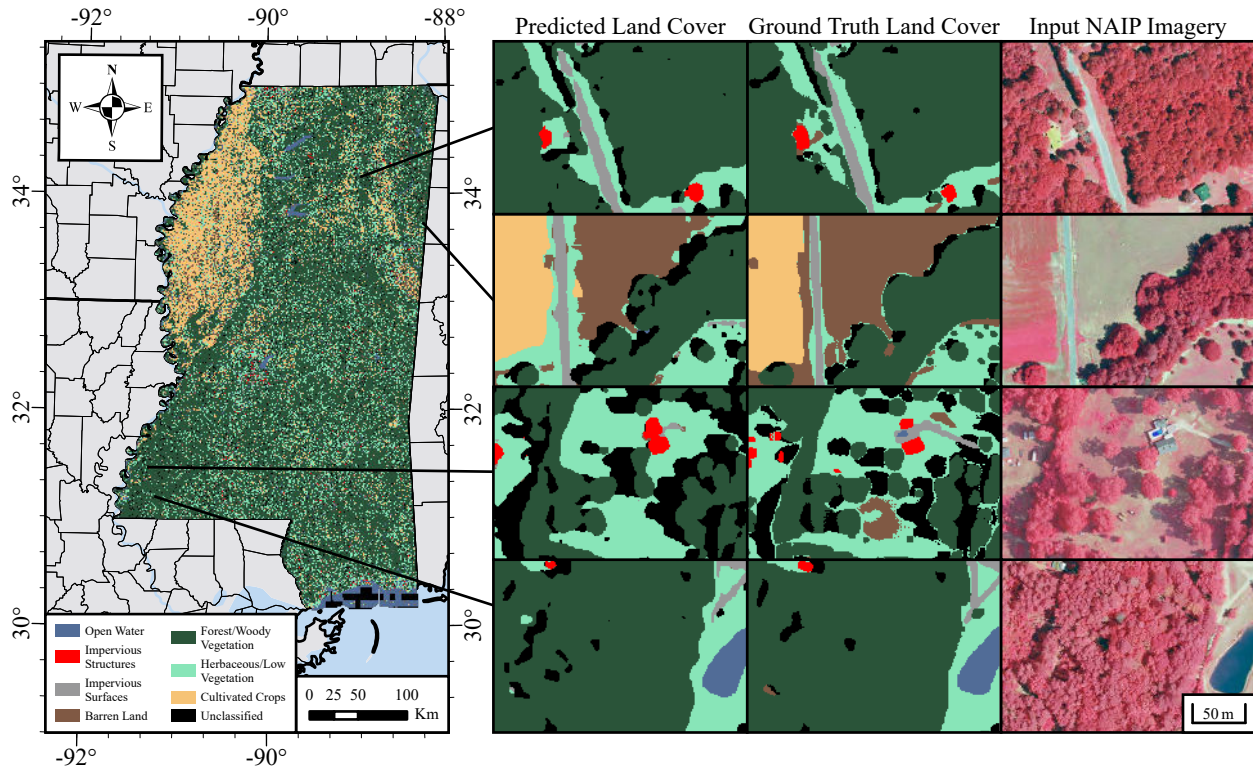


Figure 2: High-resolution land cover classification results for the Mississippi dataset (left) alongside the ground truth labels (middle) and the false-color NAIP imagery used as input (right).

Table 4: Confusion matrix for the Mississippi land cover classification results.

Predicted Label True Label	Open Water	Impervious Structures	Impervious Surfaces	Barren Land	Tree Canopy/ Woody Vegetation	Low Vegetation/ Herbaceous	Cultivated Crops	Unclassified	Producer's Accuracy
Open Water	7154	344	41	45	16	141	0	1066	81.23%
Impervious Structures	13	20776	2041	1335	115	4891	0	1708	67.28%
Impervious Surfaces	19	2590	124159	9058	1722	14407	13	3225	80.00%
Barren Land	298	1964	21482	130287	2798	100670	24151	4008	45.61%
Tree Canopy/ Woody Vegetation	924	244	1403	1064	2336744	94750	26166	101005	91.20%
Low Vegetation/ Herbaceous	469	2596	24339	57498	58379	851272	136447	29850	73.33%
Cultivated Crops	0	0	171	3079	6685	48908	148924	603	54.73%
Unclassified	208	940	3546	384	14128	8236	106	147915	62.64%
User's Accuracy	78.75%	70.54%	70.07%	64.26%	96.53%	75.78%	44.34%	51.11%	Overall Accuracy 82.11%

Table 5: Per-class F1 scores and metrics for Mississippi land cover classification results.

Class	F1 Score (%)	Support
Open Water	79.97	8,807
Impervious (raised structures)	68.87	30,879
Impervious (paved surfaces)	74.71	155,193
Barren Land	53.35	285,658
Tree Canopy/Woody Vegetation	93.79	2,562,300
Low Vegetation/Herbaceous	74.54	1,160,850
Cultivated Crops	54.73	208,370
Unclassified	63.64	175,463
Total Support		4,587,520
Macro Metrics		
User's Accuracy (%)		68.93
Producer's Accuracy (%)		74.30
F1 Score (%)		70.45
Overall Metrics		
Accuracy (%)		82.12
User's Accuracy (%)		84.06
Producer's Accuracy (%)		82.12
F1 Score (%)		82.53
Cohen's κ		0.711

In conclusion, this research demonstrates that a multi-stage pre-training approach significantly enhances land cover classification performance under label-scarce conditions. Early results show

that the proposed approach is an effective strategy for producing high spatial resolution land cover maps with limited labeled data. Beyond the immediate applications of the high spatial resolution land cover map for the state of Mississippi, this research has broader implications for land cover classification due to the unique approach to pre-training that opens the door for the development of more label efficient deep learning models. The proposed approach has the potential to enable high spatial resolution land cover classification at scale while minimizing the need for human annotation to produce ground truth data, paving the way for high resolution analyses quickly and efficiently. This is particularly important for situations where high spatial resolution land cover data for a region is necessary but existing datasets are scarce, outdated, or non-existent.

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